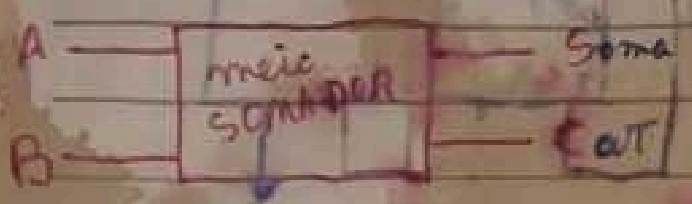
Revisão ED 2

Somadores

★ Meio Somador

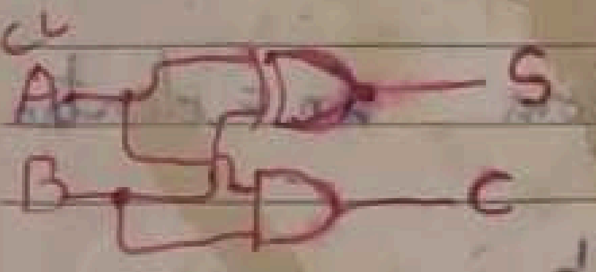


mint	Entrada A	Entrada B	Saída Soma	Saída Carry	
	0	0	0	0	
$\bar{A}B$	0	1	1	0	
$A\bar{B}$	1	0	1	0	
	1	1	0	1	AB

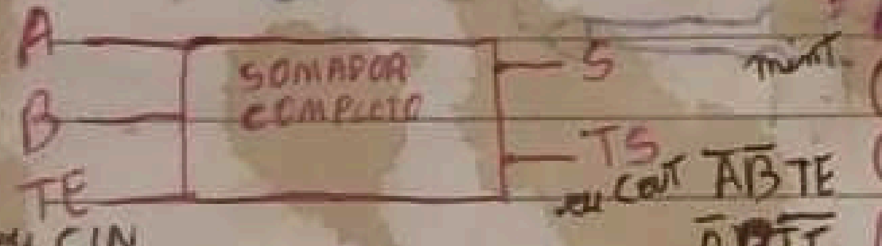
mint BOOLEANA

$S = \bar{A}B + A\bar{B}$

$S = A \oplus B \qquad C = AB$



★ Somador Completo



mint	ENTRADA A	ENTRADA B	CARRY ENTRADA (TE)	SAÍDA Soma	SAÍDA Carry Saída (TS)	
	0	0	0	0	0	
$\bar{A}\bar{B}TE$	0	0	1	1	0	
$\bar{A}B\bar{TE}$	0	1	0	1	0	
	0	1	1	0	1	$\bar{A}BTE$
$A\bar{B}\bar{TE}$	1	0	0	1	0	
	1	0	1	0	1	$A\bar{B}TE$
	1	1	0	0	1	$AB\bar{TE}$
$ABTE$	1	1	1	1	1	$ABTE$

Karnaugh S

	$\bar{B}$ $\bar{TE}$	$\bar{B}$ TE	B TE	B $\bar{TE}$
$\bar{A}$	0	1	1	1
A	1	0	1	0

$S = A\bar{B}\bar{TE} + \bar{A}\bar{B}TE + ABTE + \bar{A}B\bar{TE}$

$S = A(\bar{B}\bar{TE} + BTE) + \bar{A}(\bar{B}TE + B\bar{TE})$

$S = A(\overline{B \oplus TE}) + \bar{A}(B \oplus TE)$

$S = A \oplus B \oplus TE$

Karnaugh TS

	$\bar{B}$ $\bar{TE}$	$\bar{B}$ TE	B TE	B $\bar{TE}$
$\bar{A}$	0	0	1	0
A	0	1	1	1

$TS = ATE + BTE + AB$

$TS = TE(A+B) + AB$

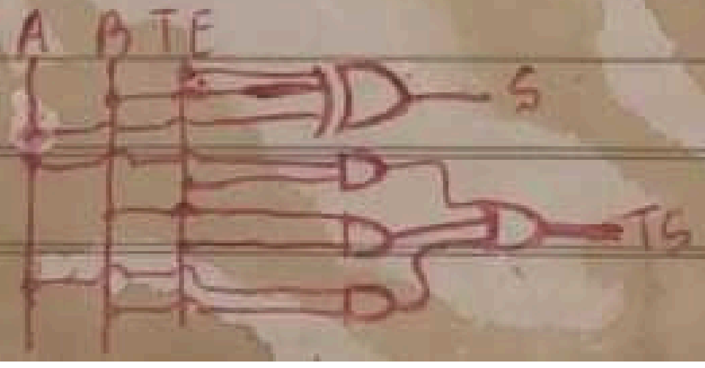
ou

$TS = (A \oplus B)TE + AB$

COUT

Obs: Se tab. Karnaugh for N será os termos com ou exclusivo / XOR " $\oplus$ "

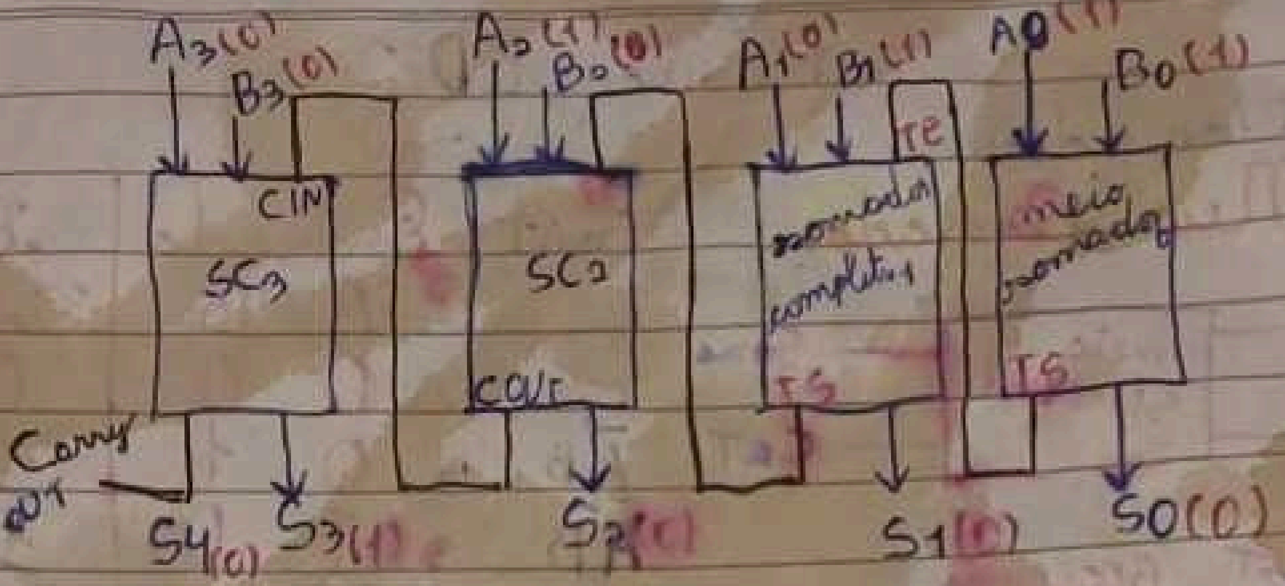
EXOR



A = 4 bits
B = 4 bits

	SC	SC	SC	MS
	A_3	A_2	A_1	A_0
+	B_3	B_2	B_1	B_0
S_4	S_3	S_2	S_1	S_0

↑ 5º bit




```
        1 1 1
A       0 1 0 1
+ B   + 0 0 1 1
        1 0 0 0
```

TS = transporte de saída
TE = transporte de entrada

Obs.: out = fora   in = dentro

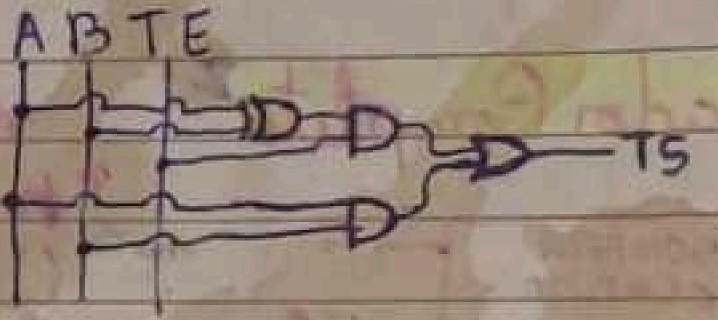
Booleana SC^2 para TS

$TS = \bar{A}BTE + A\bar{B}TE + AB\overline{TE} + ABTE$

$TE(\bar{A}B + A\bar{B}) + AB(\overline{TE} + TE)$

$TE(A \oplus B) + AB(1)$

CL

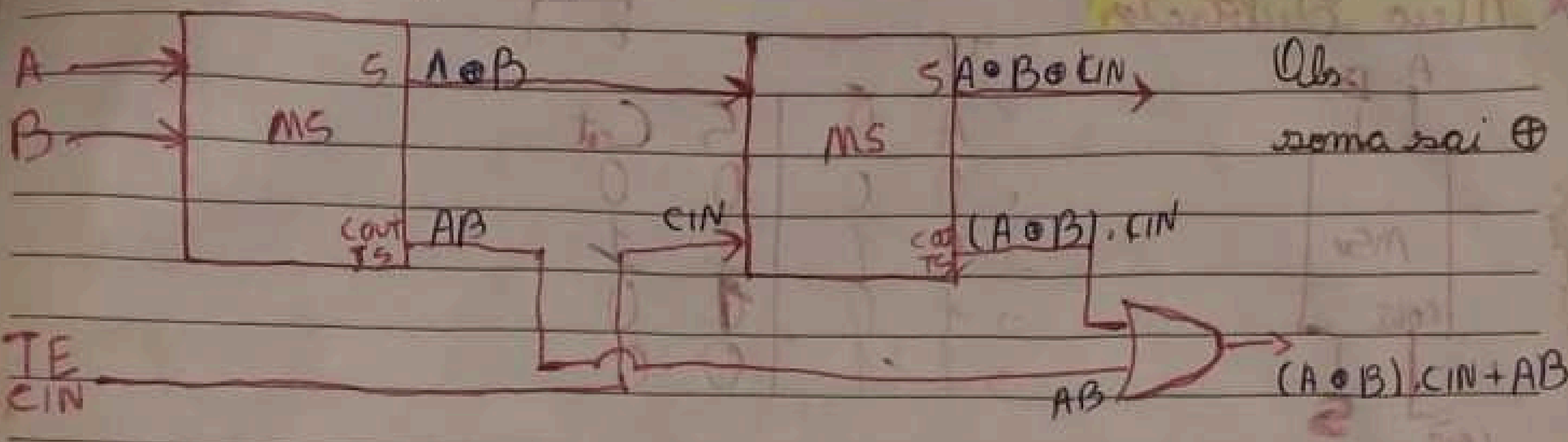


Obs para Booleana

$\bar{A}(X) + A\overline{(X)} \Rightarrow A \oplus X$

$\bar{A}\bar{X} + AX \Rightarrow \overline{A \oplus X}$ ou $A \odot X$

Somador completo por meio de meio somadores

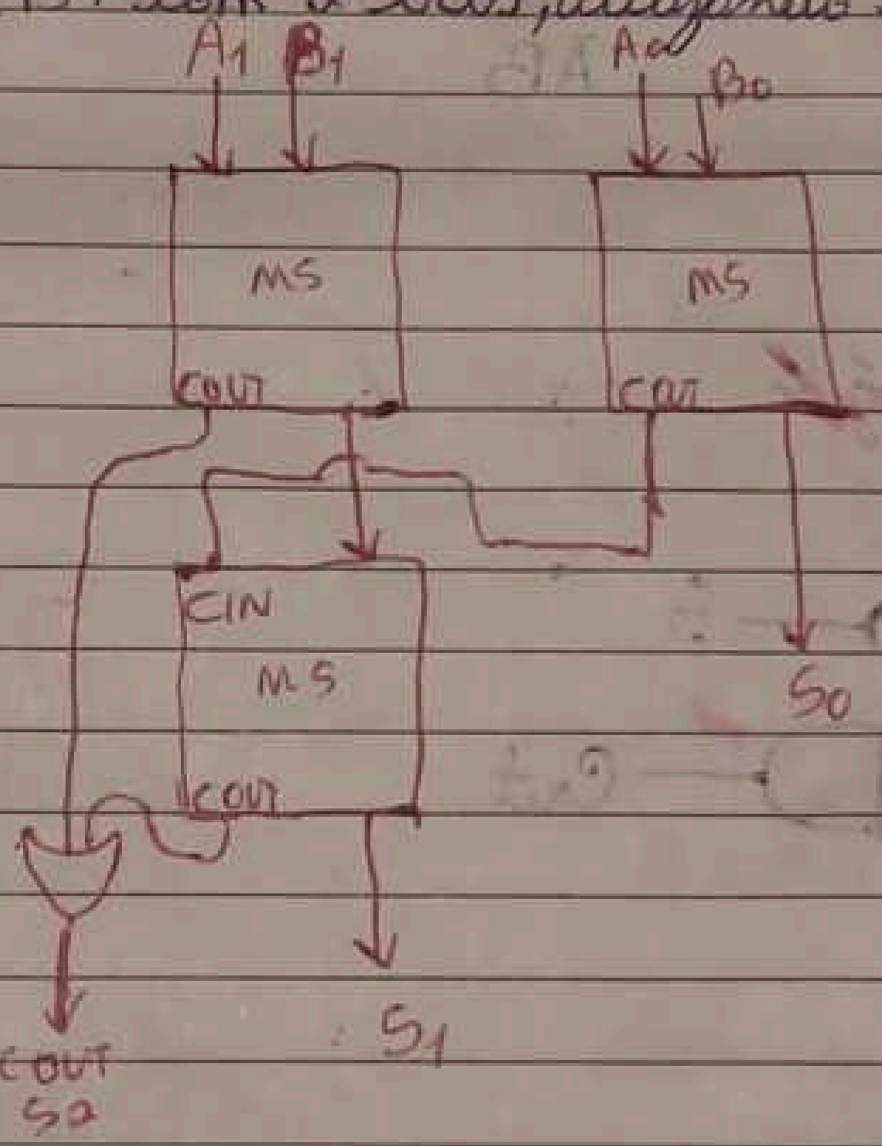


Obs: soma sai ⊕

Ex. 2 números (A e B) com 2 bits, utilizando apenas meio-somadores.

SC => 2MS   MS

A ->	A_1	A_0	
B ->	B_1	B_0	
S_2	S_1	S_0	



Se fosse 2 números (A, B) com 5 bits, utilizando apenas meio somadores

FORMULA ED

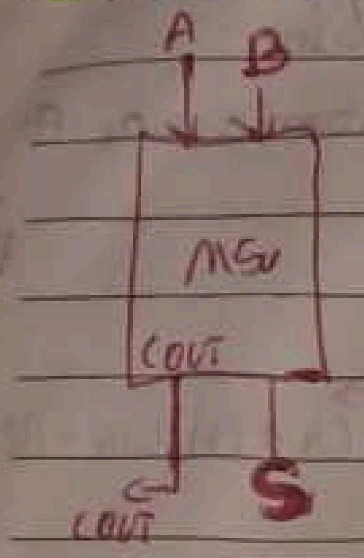
Seria 9 meio somadores   BITS-1 = NMS

Portas OR: 4   qMS = (NMS.2) + 1

qOR = Bits - 1

Subtratores

★ Meio Subtrator



TV

	A	B	S	Cout
0	0	0	0	0
1	0	1	1	1
2	1	0	1	0
3	1	1	0	0

Booleana S

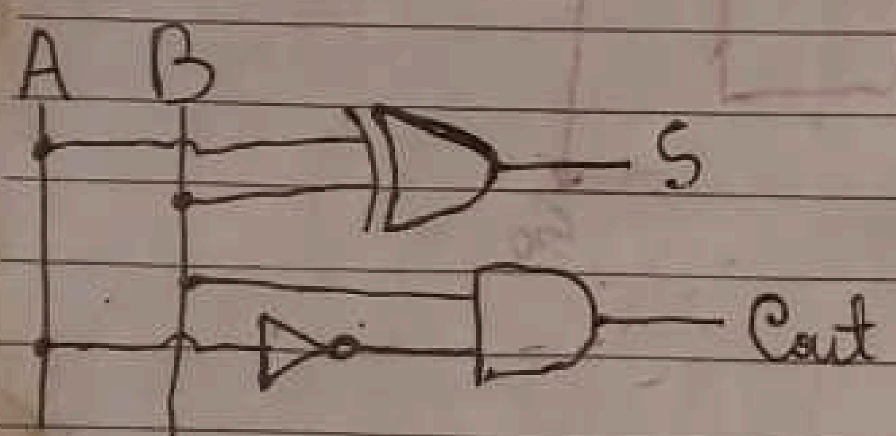
$\bar{A}B + A\bar{B}$

$A \oplus B$

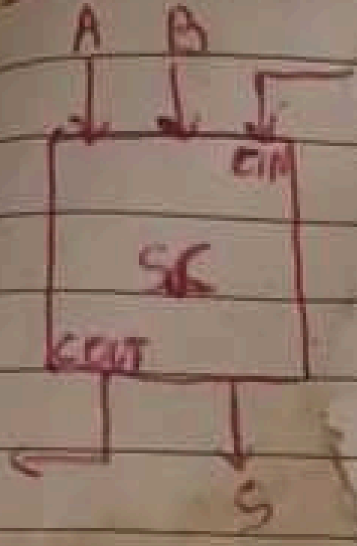
Booleana Cout TS

$\bar{A}B$

Circuito Lógico



Subtrator Completo



TV 2^3

	A	B	C_{in}	S	C_{out}
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

Booleana S

$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$

$S = \bar{A}(\bar{B}C_{in} + B\bar{C}_{in}) + A(\bar{B}\bar{C}_{in} + BC_{in})$

$S = \bar{A}(B \oplus C_{in}) + A(\overline{B \oplus C_{in}})$

$S = A \oplus B \oplus C_{in}$ //

Booleana Cout

$\bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + \bar{A}BC_{in} + ABC_{in}$

$C_{in}(\bar{A}\bar{B} + AB) + \bar{A}B(\bar{C}_{in} + C_{in})$

$C_{in}(A \odot B) + \bar{A}B$ //

Booleana Cout 2.0

$\bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + \bar{A}BC_{in} + ABC_{in}$

$\bar{A}(\bar{B}C_{in} + B\bar{C}_{in} + BC_{in}) + ABC_{in}$

$\bar{A}(\bar{B}C_{in} + B(\bar{C}_{in} + C_{in})) + ABC_{in}$

$\bar{A}C_{in} + B + ABC_{in}$

→ desnecessário pois tendo mais de dois termos é só utilizar Karnaugh

S

	$\bar{B}$	$\bar{B}$	B	B
$\bar{A}$	0	1	0	1
A	1	0	1	0
	$\bar{C}_{in}$	C_{in}	C_{in}	$\bar{C}_{in}$

$S = A \oplus B \oplus C_{in}$ //

Cout

	$\bar{B}$	$\bar{B}$	B	B
$\bar{A}$	0	1	1	1
A	0	0	1	0
	$\bar{C}_{in}$	C_{in}	C_{in}	$\bar{C}_{in}$

$C_{out} = \bar{A}C_{in} + BC_{in} + \bar{A}B$ //

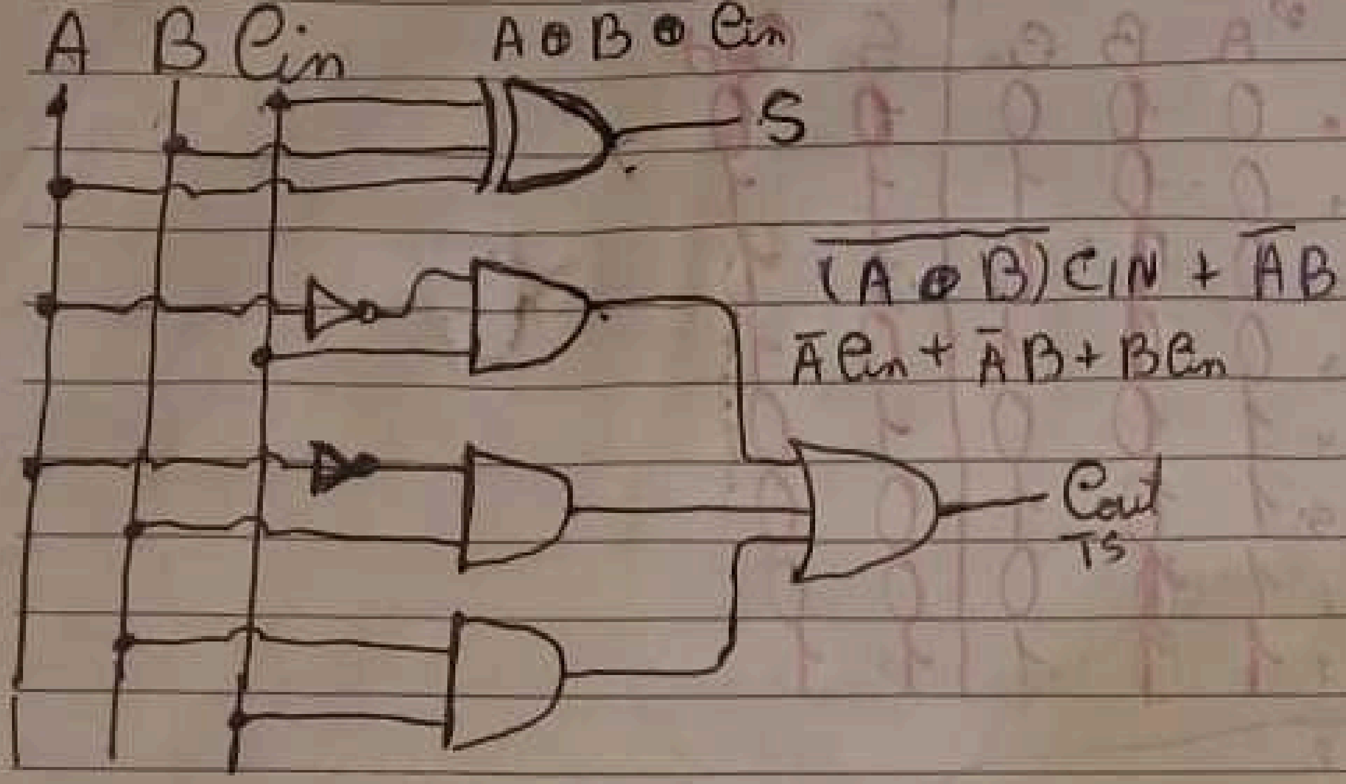
$\bar{A}(C_{in} + B) + BC_{in}$

$C_{in}(\bar{A} + B) +$

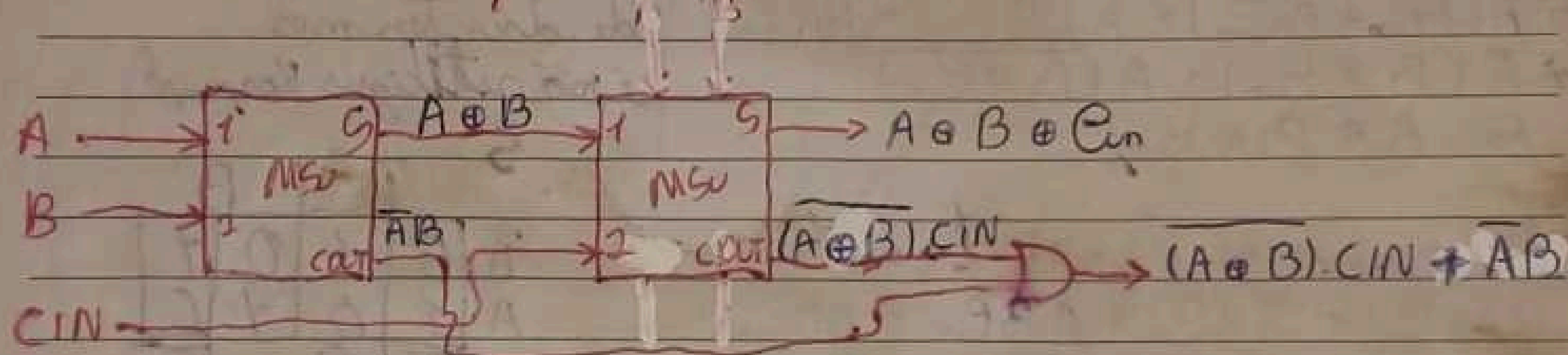
$S = A \oplus B \oplus C_{in}$

$COUT = (A \oplus B)C_{in} + \bar{A}B$

Circuito Lógico



Subtrator completo com meio subtrator

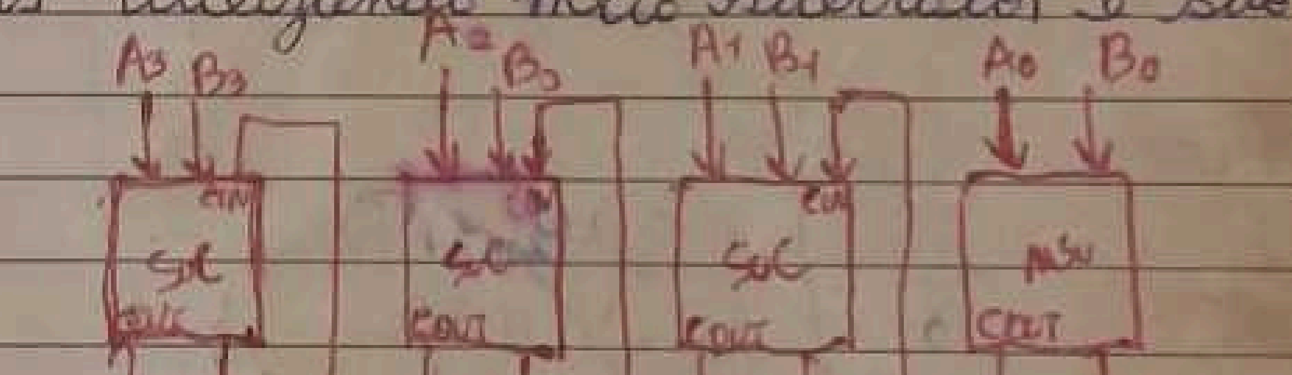


Ex: A e B com 4 bits utilizando meio subtrator e subtrator completo

$A_3\ B_2\ A_1\ A_0$

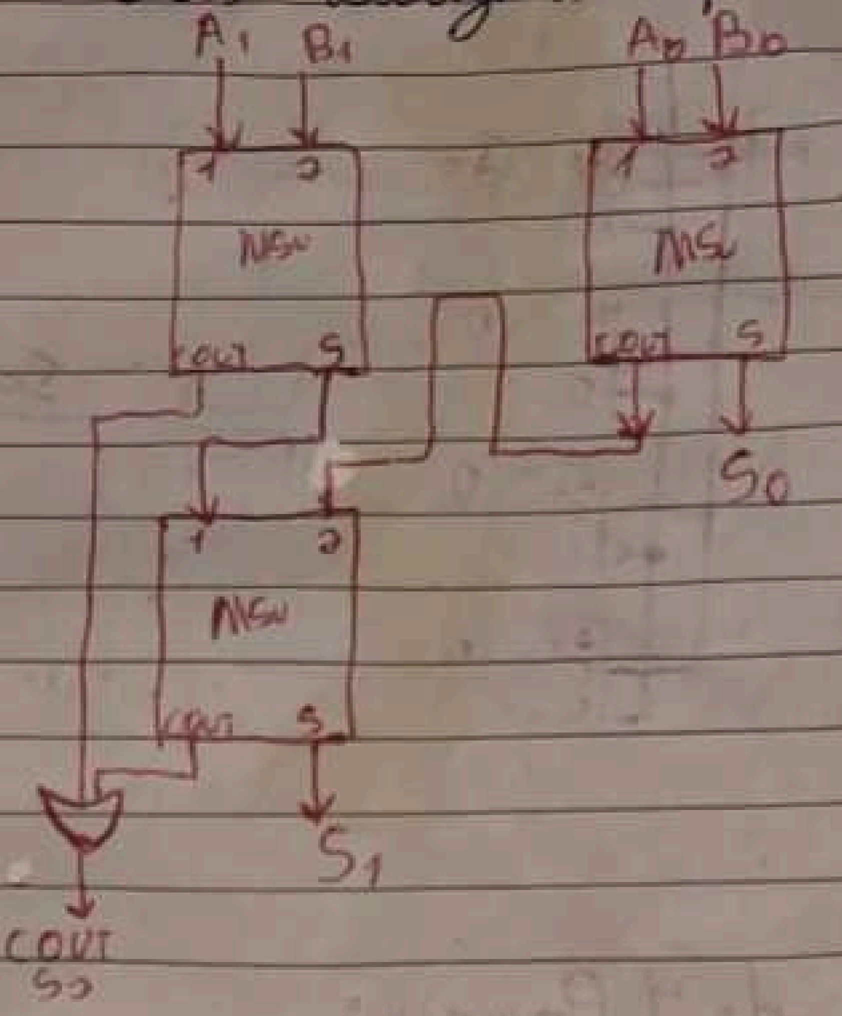
$-\ B_3\ B_2\ B_1\ B_0$

$S_4\ S_3\ S_2\ S_1\ S_0$

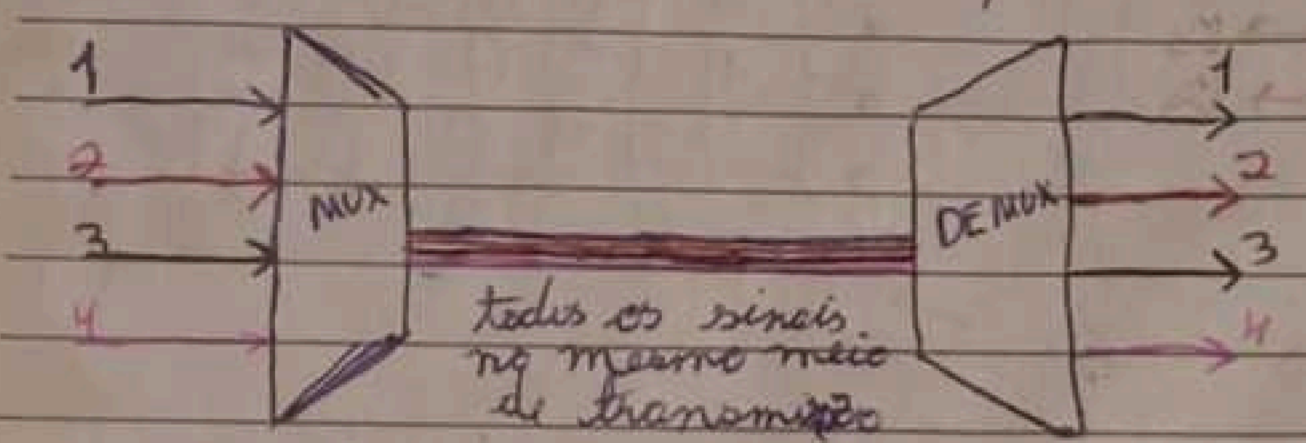


2 números (A e B) com 2 bits utilizando apenas meio subtratores

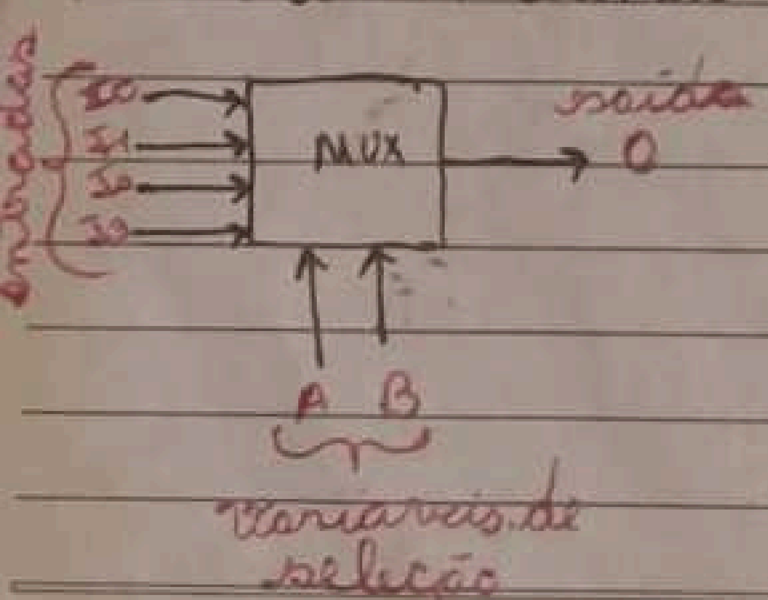
$A_1 \; A_0$
$- B_1 \; B_0$
$S_2 \; S_1 \; S_0$



★ Multiplex - Demultiplex



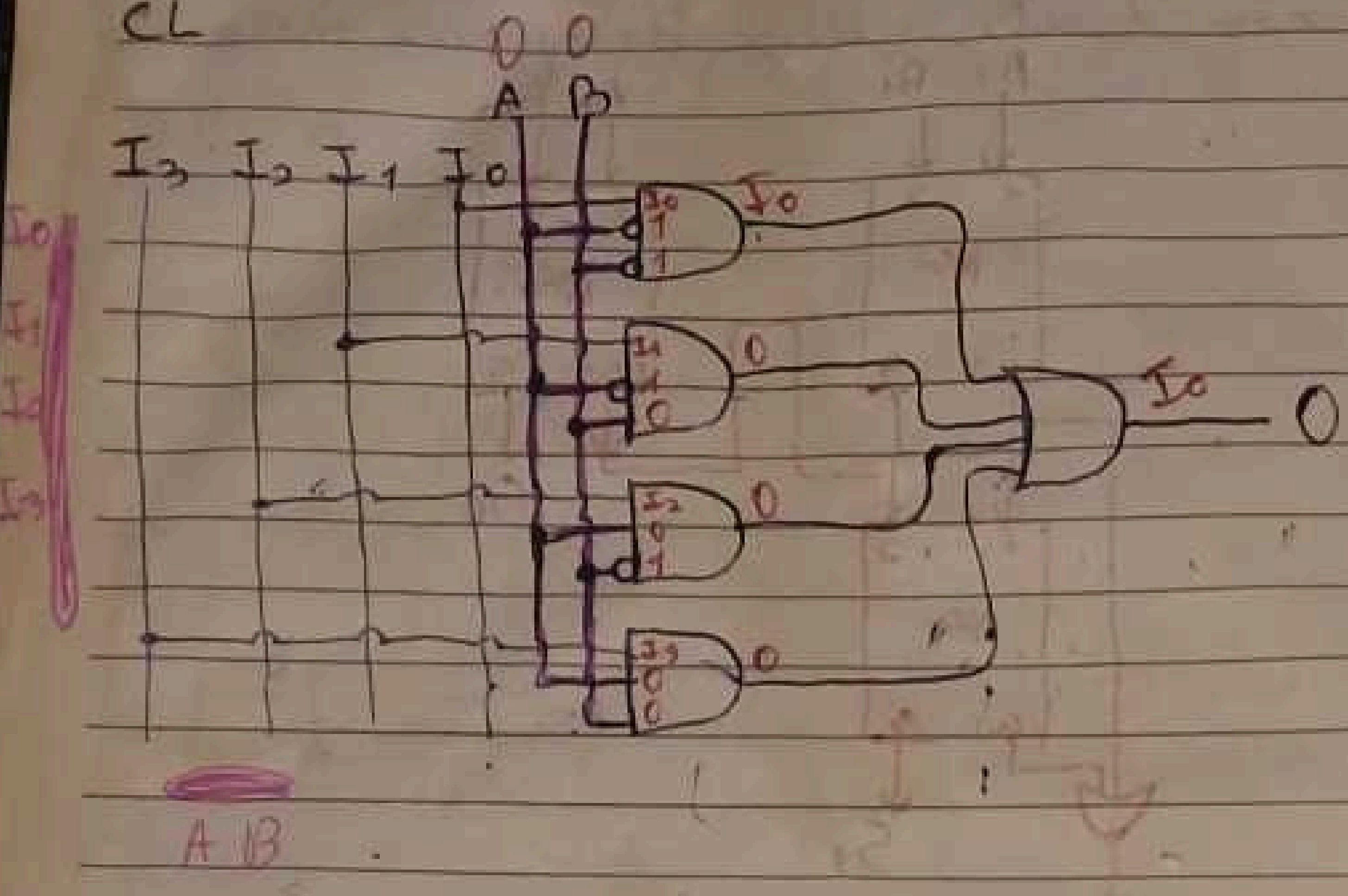
Mux de 4 canais



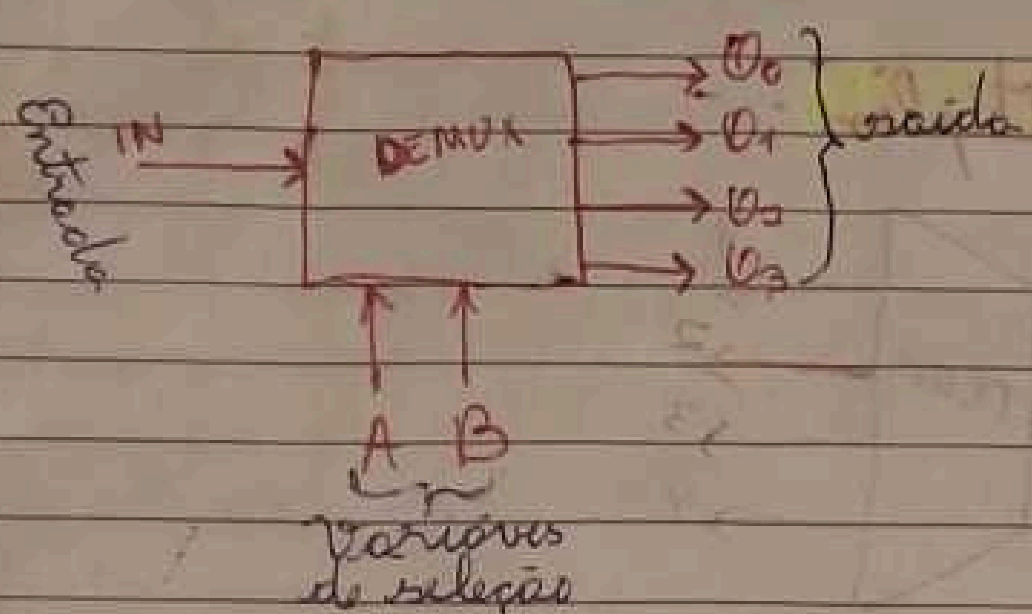
TV

	A	B	O (output)
$\bar{A}\bar{B}$	0	0	I_0
$\bar{A}B$	0	1	I_1
$A\bar{B}$	1	0	I_2
AB	1	1	I_3

CL



★ DEMUX de 4 Canais:



TV

A	B	O_0	O_1	O_2	O_3
0	0	IN	0	0	0
0	1	0	IN	0	0
1	0	0	0	IN	0
1	1	0	0	0	IN

CL